

[5460]-192

T.E.(Information Technology)
THEORY OF COMPUTATION
(2012 Pattern) (End Semester)

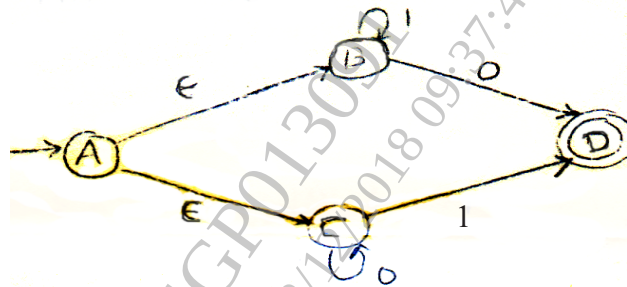
Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Neat diagram must be drawn wherever necessary.
- 2) Assume suitable data if necessary.

- Q1)** a) Write formal definition of Moore & Mealy Machine with suitable examples. Compare them. [5]
- b) Convert the following NFA ϵ into NFA without ϵ -moves. [5]



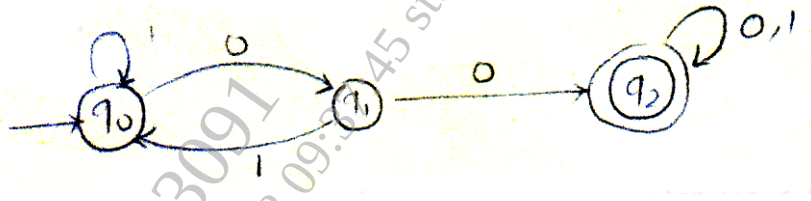
OR

- Q2)** a) Write a short note on each of the following: [5]
- i) DFA
 - ii) Relative powers of NFA and DFA.
- b) Construct NFA for the following: [5]
- i) $L = \{x \mid \Sigma = \{a, b\}; 'x' \text{ contains any number of } a\text{'s followed by at least one } b\}$
 - ii) $L = \{x \mid \Sigma = \{0, 1\}; x \text{ starts with '1' and } |x| \text{ is divisible by } 3\}$
- Q3)** a) Find regular expression for the following over $\Sigma = \{a, b\}$ [5]
- i) The language that contains strings with even number of a's
 - ii) The language that contains strings without any double letter.
- b) Show that the following language is non regular using pumping lemma.
 $L = \{ww \mid w \in (0, 1)^*\}$ [5]

P.T.O.

OR

- Q4) a)** Find the regular expression for the given DFA. [5]



- b) Express the following grammars using CNF.

$S \rightarrow bA|aB$

$A \rightarrow bAA|as|a$

$B \rightarrow aBB|bs|b$

[5]

- Q5) a)** Let $L = \{a^n b^n c^m d^m | n, m \geq 1\}$ Design a PDA that accepts L . Simulate PDA for a string. [8]

- b) Define context free language. Explain closure properties of CFL in brief. [8]

OR

- Q6) a)** Design a post machine for well formedness of parenthesis. [8]

- b) Define PDA and Post machine with suitable example. Compare them. [8]

- Q7) a)** Design a turing Machine accepting the following language. [10]

$L = \{a^i b^j | i < j\}$

Simulate TM for "aaabbbb"

- b) Write a short note on,

i) Universal TM

ii) Multi-tape TM

[8]

OR

- Q8) a)** Design a Turing machine for 2'S complement of a given binary number. Simulate TM for "1101011". [10]

- b) Define the following with suitable example.

i) Post correspondence problem

ii) Non-Deterministic TM.

[8]

Q9) a) Show that for two recursive languages L_1 & L_2 , each of the following language is recursive. [8]

i) $L_1 \cup L_2$

ii) $L_1 \cap L_2$

b) Write a short note on the following: [8]

i) Turing reducibility.

ii) Decidability.

OR

Q10) a) Prove that the halting problem of Turing Machine is unsolvable. [8]

b) Explain recursive Language and recursively enumerable language with suitable example. [8]

